R workshop part 3 - Data transformation

Jan Gačnik

Prepared: 2026-07-31

Data transformation in R

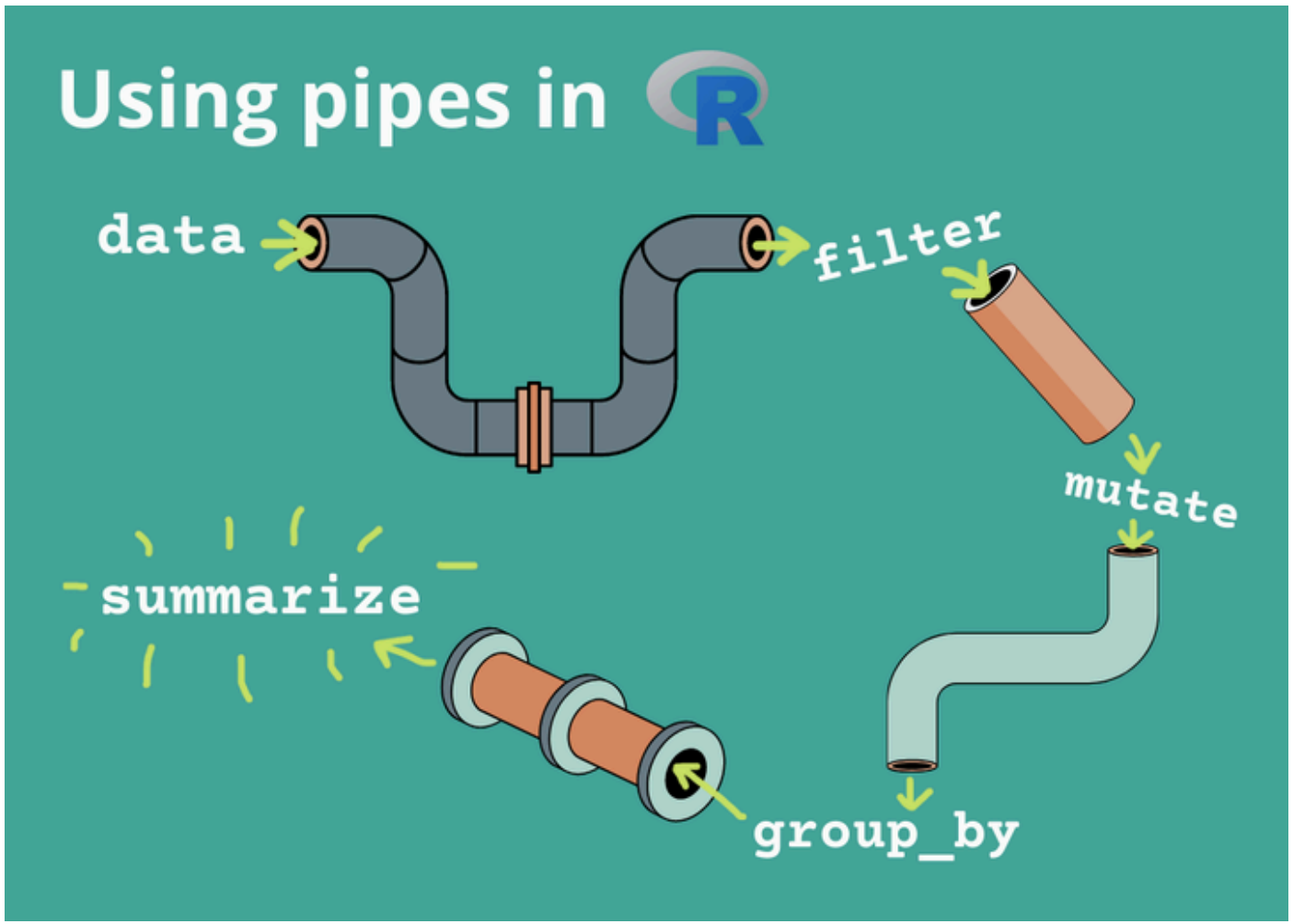
Data visualization is a great way to explore data, but the data is rarely in exact form as needed for making graphs.

In this presentation, we will learn how to:

- Filter the data
- Transform existing columns and create new ones
- Group data and make grouped summary statistics
- How to make data pipelines

Data transformation in R

Illustration of the data transformation process



Source: Celina Cheng, *R for Ecology*

Data transformation in R

We are staying in the `tidyverse` environment using the sub-package

`dplyr` 

Benefits:

- Human-readable syntax using predictable English verbs
- Intuitive piping, code as a sequence of steps
- Option to translate dplyr syntax to SQL query for databases

Cons:

- Slower than some other packages, can't handle giant data (>1 million rows)
- Functions keep improving and updating - old code may not work

Data transformation in R

Broadly, first functions studied today can be classified into these families:

- Functions that operate on **rows** of a data frame
- Functions that operate on **columns** of a data frame

1) Functions that operate on ROWS

`filter()`

Subsets a data frame, retaining all rows that satisfy your conditions.

First argument: data frame, second argument: logical test

```
1  penguins_filtered <- filter(penguins, island == "Dream")
```

Logical operator	Meaning
`<`	Less than
`<=`	Less than or equal to
`>`	More than
`>=`	More than or equal to
`==`	Equal to
`!=`	Not equal to
`A | B`	A or B
`A & B`	A and B

`filter()`

Subsets a data frame, retaining all rows that satisfy your conditions.

First argument: data frame, second argument: logical test

```
penguins_filtered <- filter(penguins, island == "Dream")
penguins_filtered_2 <- filter(penguins, flipper_len >= 190)
```

Logical operator	Meaning
`<`	Less than
`<=`	Less than or equal to
`>`	More than
`>=`	More than or equal to
`==`	Equal to
`!=`	Not equal to
`A | B`	A or B
`A & B`	A and B

`filter()`

Subsets a data frame, retaining all rows that satisfy your conditions.

First argument: data frame, second argument: logical test

```
penguins_filtered <- filter(penguins, island == "Dream")
penguins_filtered_2 <- filter(penguins, flipper_len >= 190)
penguins_filtered_3 <- filter(penguins, island == "Dream" & flipper_len >= 190)
```

Logical operator	Meaning
`<`	Less than
`<=`	Less than or equal to
`>`	More than
`>=`	More than or equal to
`==`	Equal to
`!=`	Not equal to
`A | B`	A or B
`A & B`	A and B

`filter()`

Subsets a data frame, retaining all rows that satisfy your conditions.

First argument: data frame, second argument: logical test

- Writing many `|` and `==` can get tedious:

```
1 penguins_filtered_4 <- filter(penguins, year == 2007 | year == 2008)
```

- Useful shortcut to do the same as above: `%in%`

```
1 penguins_filtered_4 <- filter(penguins, year %in% c(2007, 2008))
```

`filter()`

Subsets a data frame, retaining all rows that satisfy your conditions.

First argument: data frame, second argument: logical test

Common mistakes:

- Using `=` instead of `==`:

```
1 filter(penguins, year = 2007)
```

```
Error in `filter()`:
! We detected a named input.
ℹ This usually means that you've used `=` instead of `==`.
ℹ Did you mean `year == 2007`?
```

`filter()`

Subsets a data frame, retaining all rows that satisfy your conditions.

First argument: data frame, second argument: logical test

Common mistakes:

- Using “OR” statements like you would in English (will result in “silent error”):

```
1 filter(penguins, year == 2007 | 2008)
```

This does not give a direct error, but does not evaluate the logic the way you want.

	species	island	bill_len	bill_dep	flipper_len	body_mass	sex	year
1	Adelie	Torgersen	39.1	18.7	181	3750	male	2007
2	Adelie	Torgersen	39.5	17.4	186	3800	female	2007
3	Adelie	Torgersen	40.3	18.0	195	3250	female	2007
4	Adelie	Torgersen	NA	NA	NA	NA	<NA>	2007
5	Adelie	Torgersen	36.7	19.3	193	3450	female	2007
6	Adelie	Torgersen	39.3	20.6	190	3650	male	2007
7	Adelie	Torgersen	38.9	17.8	181	3625	female	2007
8	Adelie	Torgersen	39.2	19.6	195	4675	male	2007

```
9      Adelie Torgersen     34.1     18.1          193       3475   <NA> 2007
10     Adelie Torgersen     42.0     20.2          190       4250   <NA> 2007
11     Adelie Torgersen     37.8     17.1          186       3300   <NA> 2007
12     Adelie Torgersen     37.8     17.3          180       3700   <NA> 2007
```

distinct()

Keeps only unique/distinct rows from a data frame

First argument: data frame, next arguments: variables from the data frame for which a distinct combination is wanted

```
1 distinct(penguins, species, island)
```

```
    species    island
1    Adelie Torgersen
2    Adelie    Biscoe
3    Adelie     Dream
4    Gentoo    Biscoe
5 Chinstrap     Dream
```

`count()`

Similar to `distinct()`, but rather than showing you unique instances it counts the number of occurrences

First argument: data frame, next arguments: variables whose unique combinations are counted

```
1 count(penguins, species, island)
```

```
    species    island   n
1    Adelie    Biscoe  44
2    Adelie     Dream  56
3    Adelie Torgersen  52
4 Chinstrap     Dream  68
5    Gentoo    Biscoe 124
```

- Optionally, you can arrange the results in descending order using `sort = TRUE` argument.

`slice()`

Allows you to select, remove, and duplicate rows based on their (integer) location. E.g., rows 1-5, rows 20-last row, etc.

First argument: data frame, second argument: integers of the row location.

- **positive** integers indicate you want to keep rows, **negative** integers indicate you want to remove rows

```
1 slice(penguins, 2:4)
```

```
  species    island bill_len bill_dep flipper_len body_mass    sex year
1  Adelie Torgersen     39.5     17.4          186      3800 female 2007
2  Adelie Torgersen     40.3     18.0          195      3250 female 2007
3  Adelie Torgersen       NA       NA           NA        NA   <NA> 2007
```

`slice()`

Allows you to select, remove, and duplicate rows based on their (integer) location. E.g., rows 1-5, rows 20-last row, etc.

First argument: data frame, second argument: integers of the row location.

- **positive** integers indicate you want to keep rows, **negative** integers indicate you want to remove rows

```
1 slice(penguins, -(2:4))
```

```
     species    island bill_len bill_dep flipper_len body_mass    sex year
1     Adelie Torgersen     39.1     18.7          181      3750   male 2007
2     Adelie Torgersen     36.7     19.3          193      3450 female 2007
3     Adelie Torgersen     39.3     20.6          190      3650   male 2007
4     Adelie Torgersen     38.9     17.8          181      3625 female 2007
5     Adelie Torgersen     39.2     19.6          195      4675   male 2007
6     Adelie Torgersen     34.1     18.1          193      3475   <NA> 2007
7     Adelie Torgersen     42.0     20.2          190      4250   <NA> 2007
```

```
8      Adelie Torgersen     37.8     17.1          186       3300   <NA> 2007
9      Adelie Torgersen     37.8     17.3          180       3700   <NA> 2007
10     Adelie Torgersen     41.1     17.6          182       3200 female 2007
11     Adelie Torgersen     38.6     21.2          191       3800   male 2007
12     Adelie Torgersen     34.6     21.1          198       4400   male 2007
```

2) Functions that operate on COLUMNS

`mutate()`

Creates new or modifies existing variables in the object

First argument: data frame, second argument: name-value pair

```
1 penguins_mutated <- mutate(penguins, ratio = flipper_len / body_mass)
```

```
  species    island bill_len bill_dep flipper_len body_mass    sex year      ratio
1  Adelie Torgersen     39.1     18.7         181      3750   male 2007 0.04826667
2  Adelie Torgersen     39.5     17.4         186      3800 female 2007 0.04894737
3  Adelie Torgersen     40.3     18.0         195      3250 female 2007 0.06000000
4  Adelie Torgersen       NA       NA          NA        NA   <NA> 2007         NA
5  Adelie Torgersen     36.7     19.3         193      3450 female 2007 0.05594203
6  Adelie Torgersen     39.3     20.6         190      3650   male 2007 0.05205479
```

mutate()

Creates new or modifies existing variables in the object

First argument: data frame, second argument: name-value pair

```
1 penguins_mutated <- mutate(penguins, ratio = flipper_len / body_mass)
2 penguins_mutated_2 <- mutate(penguins, body_mass_kg = body_mass / 1000)
```

```
  species    island bill_len bill_dep flipper_len body_mass    sex year body_mass_kg
1  Adelie Torgersen     39.1     18.7         181      3750   male 2007         3.75
2  Adelie Torgersen     39.5     17.4         186      3800 female 2007         3.80
3  Adelie Torgersen     40.3     18.0         195      3250 female 2007         3.25
4  Adelie Torgersen       NA       NA          NA        NA   <NA> 2007           NA
5  Adelie Torgersen     36.7     19.3         193      3450 female 2007         3.45
6  Adelie Torgersen     39.3     20.6         190      3650   male 2007         3.65
```

mutate()

Creates new or modifies existing variables in the object

First argument: data frame, second argument: name-value pair

```
penguins_mutated <- mutate(penguins, ratio = flipper_len / body_mass)
penguins_mutated_2 <- mutate(penguins, body_mass_kg = body_mass / 1000)
penguins_mutated_3 <- mutate(penguins, size = ifelse(body_mass > 4000, "Large", "Small"))
```

```
  species    island bill_len bill_dep flipper_len body_mass    sex year  size
1  Adelie Torgersen     39.1     18.7         181      3750   male 2007 Small
2  Adelie Torgersen     39.5     17.4         186      3800 female 2007 Small
3  Adelie Torgersen     40.3     18.0         195      3250 female 2007 Small
4  Adelie Torgersen       NA       NA          NA        NA   <NA> 2007  <NA>
5  Adelie Torgersen     36.7     19.3         193      3450 female 2007 Small
6  Adelie Torgersen     39.3     20.6         190      3650   male 2007 Small
```

select()

Select (and optionally rename) variables in a data frame

First argument: data frame, next arguments: names of columns to keep or remove

```
1 select(penguins, species, bill_len, bill_dep)
```

```
      species bill_len bill_dep
1      Adelie     39.1     18.7
2      Adelie     39.5     17.4
3      Adelie     40.3     18.0
4      Adelie       NA       NA
5      Adelie     36.7     19.3
6      Adelie     39.3     20.6
7      Adelie     38.9     17.8
8      Adelie     39.2     19.6
9      Adelie     34.1     18.1
10     Adelie     42.0     20.2
11     Adelie     37.8     17.1
12     Adelie     37.8     17.3
13     Adelie     41.1     17.6
14     Adelie     38.6     21.2
15     Adelie     34.6     21.1
```

`select()`

Select/remove variables in a data frame

First argument: data frame, next arguments: names of columns to keep or remove

- Useful functions that can be used as arguments of `select()`: `starts_with()`, `ends_with()`, and `contains()`

```
1  select(penguins, starts_with("bill"))
```

```
    bill_len bill_dep
1       39.1     18.7
2       39.5     17.4
3       40.3     18.0
4         NA       NA
5       36.7     19.3
6       39.3     20.6
7       38.9     17.8
8       39.2     19.6
9       34.1     18.1
```

`select()`

Select/remove variables (columns) in a data frame

First argument: data frame, next arguments: names of columns to keep or remove

- Useful functions that can be used as arguments of `select()`: `starts_with()`, `ends_with()`, and `contains()`

```
1 select(penguins, contains("_"))
```

- As before, putting `-` in front of the argument will remove the specified columns

rename()

Rename variables in a data frame

First argument: data frame, next arguments: “new_name” = old_name, “new_name_2” = old_name_2, …

```
1 rename(penguins, "bill_len_mm" = bill_len)
```

```
     species    island bill_len_mm bill_dep flipper_len body_mass    sex year
1     Adelie Torgersen        39.1     18.7         181      3750   male 2007
2     Adelie Torgersen        39.5     17.4         186      3800 female 2007
3     Adelie Torgersen        40.3     18.0         195      3250 female 2007
4     Adelie Torgersen          NA       NA          NA        NA   <NA> 2007
5     Adelie Torgersen        36.7     19.3         193      3450 female 2007
6     Adelie Torgersen        39.3     20.6         190      3650   male 2007
7     Adelie Torgersen        38.9     17.8         181      3625 female 2007
8     Adelie Torgersen        39.2     19.6         195      4675   male 2007
9     Adelie Torgersen        34.1     18.1         193      3475   <NA> 2007
```

3) Grouping

Group & summarise data

Function `group_by()` *silently* creates grouping of data by a specified variable

This means you will see only `# Groups` added to the data frame header, nothing will change with data itself

```
1 group_by(penguins, species)
```

```
# A tibble: 344 × 8
# Groups:   species [3]
   species island    bill_len bill_dep flipper_len body_mass sex     year
   <fct>   <fct>        <dbl>    <dbl>       <int>     <int> <fct>  <int>
 1 Adelie  Torgersen     39.1     18.7         181      3750 male    2007
 2 Adelie  Torgersen     39.5     17.4         186      3800 female  2007
 3 Adelie  Torgersen     40.3     18           195      3250 female  2007
 4 Adelie  Torgersen     NA       NA            NA        NA <NA>    2007
 5 Adelie  Torgersen     36.7     19.3         193      3450 female  2007
 6 Adelie  Torgersen     39.3     20.6         190      3650 male    2007
 7 Adelie  Torgersen     38.9     17.8         181      3625 female  2007
 8 Adelie  Torgersen     39.2     19.6         195      4675 male    2007
 9 Adelie  Torgersen     34.1     18.1         193      3475 <NA>    2007
10 Adelie  Torgersen     42       20.2         190      4250 <NA>    2007
# ℹ 334 more rows
```

Group & summarise data

Function `summarise()` then collapses many rows into single summary value *per group*, the summary statistic can be mean, median, quartiles, etc.

If data is not grouped, the function will be used on complete data.

- Let's first demonstrate the summarise on non-grouped data:

```
1  summarise(penguins, avg_len = mean(flipper_len))
```

```
  avg_len
1      NA
```

Group & summarise data

Function `summarise()` then collapses many rows into single summary value *per group*, the summary statistic can be mean, median, quartiles, etc.

If data is not grouped, the function will be used on complete data.

- That did not work, R does not know how to deal with `NA` values when applying summary functions

- Ignoring `NA` values can be specified using `na.rm = TRUE` argument:

```
1  summarise(penguins, avg_len = mean(flipper_len, na.rm = TRUE))
```

```
   avg_len
1 200.9152
```

Group & summarise data

`group_by()` and `summarise()` usually work in tandem, where data is first grouped, and then summarised by group.

- Let's pass `penguins` grouped by `species` to the `summarise()` function

```
summarise(group_by(penguins, species),
          avg_len = mean(flipper_len, na.rm = TRUE))
```

```
# A tibble: 3 × 2
  species   avg_len
  <fct>       <dbl>
1 Adelie       190.
2 Chinstrap    196.
3 Gentoo       217.
```

Code readability

Our code chunks will slowly become longer and *more nested*, or we will encounter *naming issues*. Examples:

```
# A nested code chunk
select(
  mutate(
    filter(
      penguins,
      year >= 2008
    ),
    size = ifelse(body_mass > 4000, "large", "small")
  ),
  species, year, size
)
```

```
# So many similar names!
penguins_1 <- filter(penguins, year >= 2008)
penguins_2 <- mutate(penguins_1, size = ifelse(body_mass > 4000, "large", "small"))
penguins_3 <- select(penguins_2, species, year, size)
```

Solution? **Data pipelines**

4) Pipelines

Data pipelines

Data pipelines are a way of changing multiple operations into a clear, readable sequence using the pipe operator: `%>%` (tidyverse) or `|>` (base R).

A shortcut for writing the pipe operator is `ctrl` + `shift` + `M`.

- Let's demonstrate this on the previous example:

```
summarise(group_by(penguins, species),
          avg_len = mean(flipper_len, na.rm = TRUE))
```

- Same example with pipeline approach:

```
penguins %>%
  filter(is.na(flipper_len) == FALSE)
```

Data pipelines

Data pipelines are a way of changing multiple operations into a clear, readable sequence using the pipe operator: `%>%` (tidyverse) or `|>` (base R).

A shortcut for writing the pipe operator is `ctrl` + `shift` + `M`.

- Let's demonstrate this on the previous example:

```
summarise(group_by(penguins, species),
          avg_len = mean(flipper_len, na.rm = TRUE))
```

- Same example with pipeline approach:

```
penguins %>%
  filter(is.na(flipper_len) == FALSE) %>%
  group_by(species, sex)
```

Data pipelines

Data pipelines are a way of changing multiple operations into a clear, readable sequence using the pipe operator: `%>%` (tidyverse) or `|>` (base R).

A shortcut for writing the pipe operator is `ctrl` + `shift` + `M`.

- Let's demonstrate this on the previous example:

```
summarise(group_by(penguins, species),
          avg_len = mean(flipper_len, na.rm = TRUE))
```

- Same example with pipeline approach:

```
penguins %>%
  filter(is.na(flipper_len) == FALSE) %>%
  group_by(species, sex) %>%
  summarise(avg_len = mean(flipper_len))
```

Data pipelines

- Let's try to get medians with interquartile ranges *per penguin species* and *per penguin sex*

```
1 penguins %>%
2   filter(is.na(flipper_len) == FALSE & is.na(sex) == FALSE)
```

Data pipelines

- Let's try to get medians with interquartile ranges *per penguin species* and *per penguin sex*

```
1 penguins %>%
2   filter(is.na(flipper_len) == FALSE & is.na(sex) == FALSE) %>%
3   group_by(species, sex)
```

Data pipelines

- Let's try to get medians with interquartile ranges *per penguin species* and *per penguin sex*

```
penguins %>%
  filter(is.na(flipper_len) == FALSE & is.na(sex) == FALSE) %>%
  group_by(species, sex) %>%
  summarise(median_flipper = median(flipper_len),
            q25 = quantile(flipper_len, 0.25),
            q75 = quantile(flipper_len, 0.75))
```

```
# A tibble: 6 × 5
# Groups:   species [3]
  species   sex    median_flipper   q25   q75
  <fct>     <fct>           <dbl> <dbl> <dbl>
1 Adelie    female            188   185   191
2 Adelie    male              193   189   197
3 Chinstrap female            192   187.  196.
4 Chinstrap male              200.  196   203
5 Gentoo    female            212   210   215
6 Gentoo    male              221   218   225
```

Important functions and concepts, let's revise briefly before moving to next topics